

AARAV ARTHAM

aarav.artham@utexas.edu | (469) 578-9737 | McKinney, Texas 75070 | US Citizen

EDUCATION

Bachelor of Science in Electrical and Computer Engineering,

University of Texas at Austin, Cockrell School of Engineering

05/2028 | Austin, Texas

- GPA: 3.84

WORK EXPERIENCE

Controls and Telemetry Engineer, Long Horn Racing Solar

08/2024 – Present | Austin, Texas

- Designed sensor boards to monitor ride height, humidity, temperature, strain, and irradiance across **CAN, UART, and I2C** buses, bringing **20+ strain gauges, thermocouples, black-body IR sensors**, and an **irradiance sensor** onto a shared telemetry network
- Integrated Hall effect sensors for contactless wheel speed and ride-height measurement, strain gauges for suspension sensing
- Programmed firmware logging at **50 Hz**, capturing **10,000+ data** points for post-competition analysis of suspension conditions
- Reduced vehicle mass by **~70 lb** by transitioning from a space frame to a composite chassis; designed aeroshell bulkheads in **SolidWorks** and **AutoCAD**, validating placement with **ANSYS FEA/CFD** and **MATLAB**

Undergraduate Researcher - Radionavigation Laboratory, University of Texas at Austin

08/2026 – Present | Austin, Texas

- Conducted autonomous swarm robotics research under Dr. Todd Humphreys, migrating the fleet's control stack to **ROS2**
- Contributed drone-array beamforming results that helped secure **\$15M** in research funding from Catalyst and Duke, delivering **~6–10 dB** of RF signal gain at the focal point through synchronized flight
- Designed **C++/ROS2 PID** control loops for real-time attitude and position stabilization, achieving world-first centimeter-level precision across an **8-drone** coordinated formation
- Upgrading all 8 drones to **NVIDIA Jetson Nano's** to enable edge processing for autonomous coordination

Undergraduate Researcher - SPYP Engineering Lab, Texas A&M

05/2024 – 08/2024 | College Station, Texas

- Conducted research on the bacterial growth of *Alternaria alternata* and *Aspergillus niger* to prove effectiveness of air purifier
- Designed housing for system components using **Autodesk Fusion** and **ANSYS FEA**, supporting structural integration
- Applied findings to indoor **air filtration systems**, with implications for residential and **food manufacturing environments**
- Co-authored peer-reviewed research paper and presented findings to **600+** attendees at the **SRW Research Symposium**

Web Development Intern, Rainfolutions

05/2024 – 08/2024 | Dallas, Texas

- Reduced website page load time by **25-35%** through performance optimization and routine infrastructure maintenance
- Managed **20+** job postings and executed **~20** updates per month, ensuring accurate content and consistent site functionality

PROJECT EXPERIENCE

AI-Assisted Telemetry Diagnostics Tool (Python + LLM Software)

05/2026 – 08/2026

- Building a Python pipeline to ingest **CAN-bus** and sensor telemetry logs from **10,000+** data points per test session, applying an **LLM-based** analysis layer to auto-generate plain-language diagnostic summaries and flag anomalous subsystem behavior
- Designing a lightweight retrieval layer targeting a **30-40% reduction** in manual log-review time during bench testing once deployed
- Prototyping anomaly-flagging logic aimed at surfacing subsystem faults within seconds of a bench-test run, versus manual review that currently takes hours per session

Raspberry Pi Cluster & Arduino Data Center (Embedded Systems)

05/2026 – 08/2026

- Designed a computing cluster using **5 Raspberry Pi nodes** and developed custom **C/C++ task-scheduling software**, increasing parallel workload-processing speed by approximately **40%** compared to a single-node system.
- Integrated **6 Arduino microcontrollers** to remotely control devices, execute scheduled automation routines, and report system status.
- Developed system-logging and diagnostic tools to track task execution, communication failures, and device activity, reducing troubleshooting and manual configuration time by approximately **30%**.

Closed-Loop Servo Control System (Embedded PID Control)

05/2025 – 08/2025

- Developed microcontroller firmware for a real-time **PID position-control** loop, reducing overshoot by approximately **60%** and settling time by approximately **45%** compared with open-loop behavior.
- Integrated potentiometer feedback and executed the control loop at approximately **50 Hz**; characterized rise time, steady-state error, stability, and step response tradeoffs.

Technical Skills

Embedded/Firmware: C/C++, Python, ESP32, Arduino, STM32CubeIDE, STM32CubeMX, Raspberry Pi, embedded Linux, PID control

Interfaces & Data: CAN, UART, I2C, real-time sensor sampling, data logging, networked device control, system diagnostics.

Hardware & Integration: KiCad PCB design, PCB layout and testing, sensor integration, vehicle electronics, grounding, connector reliability, bench testing

Design & Analysis: SolidWorks, ANSYS FEA, MATLAB, Autodesk Fusion, materials testing, 3D printing, machine shop, laser cutting

Software Engineering & Tools: Git/version control, LabVIEW, Python data analysis & automation, LLM-assisted diagnostics tooling

ADDITIONAL

Clubs: Long Horn Racing Solar, Institute of Electrical and Electronics Engineers, IEEE Robotics and Automation Society

Work Eligibility: U.S. Citizen and U.S. Person, eligible to work in the U.S. with no restrictions